

Darren Biskup

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EDUCATION

Columbia University

M.S./Ph.D. Mechanical Engineering (Robotics & Rehabilitation)

Columbia-Dream Sports AI Ph.D. Fellowship

New York, NY

Aug 2024 - Present

GPA: 4.11/4.33

University of Illinois, Urbana-Champaign

B.S. Mechanical Engineering, Computer Science Minor

High Honors, James Scholar Honors Program, Dean's List

Urbana, IL

May 2024

GPA: 3.95

SKILLS

Languages & Tools: C/C++, Python (PyTorch, JAX, OpenCV), C#/Unity, Linux/Bash, Git, CMake

Controls & Estimation: Convex MPC, Trajectory Optimization, Sensor Fusion (EKF/UKF), ALGLIB, OSQP

Autonomy & Simulation: MuJoCo/MJPC, pyElastica, ROS2, Predictive Safety Filters, Domain Randomization

Hardware & Systems: CAN/CAN-FD, BLDC control, Raspberry Pi, pneumatics, force plates/load cells

EXPERIENCE

Robotics And Rehabilitation (ROAR) Lab, Columbia University

Aug 2024 - Present

Graduate Research Assistant

- Authored an IEEE RA-L submission on real-time ground-reaction-force-aware convex MPC using human centroidal dynamics for underactuated cable-driven robotic standing assistance.
- Implemented the MPC as a C#/Unity QP using ALGLIB QuickQP, reducing dynamic tracking error by 24% and mechanical work by 29% versus impedance control; [Code].
- Integrated a hierarchical real-time control stack with host-level MPC, asynchronous TCP tension-command streaming, 1-kHz load-cell feedback, and CAN-controlled EPOS motor drives, incorporating Vicon/HTC Vive motion capture and force-plate sensing.

Mehta Research Group/GazzolaLab, UIUC

Jan 2023 - May 2024

Undergraduate Research Assistant - Soft Robotics

- Built an OpenCV/scikit-image calibration pipeline to extract accurate 3D marker coordinates from a custom spatial grid for robotic-arm pose tracking.
- Developed Python control scripts synchronizing a physical joystick-actuated soft robotic arm with its *pyElastica* digital twin to evaluate sim-to-real dynamic fidelity.

Skydio

May 2022 - Aug 2022

Product Design Engineering Intern

- Designed navigation-camera mounting features for the Skydio X10 and coordinated with overseas vendors to implement design changes while meeting injection-molding and lead-time constraints.
- Collaborated with manufacturing and autonomy teams on design-for-manufacturing (DFM) and spatial hardware layout for the next-generation autonomous performance quadcopter.

Lucid Motors

May 2021 - Aug 2021

Mechanical Engineering Intern, High Voltage

- Conducted data-driven thermal and efficiency analysis on high-voltage battery pack assemblies to optimize hardware performance and range.

ROBOTICS & AUTONOMY PROJECTS

Predictive Safety Filter for MuJoCo Quadrotor Flight

Python, JAX, MuJoCo, SciPy

- Implemented a finite-horizon predictive safety filter for online collision avoidance of a MuJoCo quadrotor in constrained workspaces with modeled obstacles. Optimizes motor commands over forward-simulated trajectories to minimize deviation from requested inputs while maintaining collision-free motion.
- Used JAX's just-in-time (JIT) compilation and automatic differentiation to accelerate repeated nonlinear physics rollouts and SLSQP cost and constraint evaluations.

C++ MuJoCo GRF Data-Generation Pipeline

MuJoCo, MJPC, C++, CMake

- Building a C++17 MuJoCo/MJPC data factory that replays motion-capture-derived anatomical marker trajectories and records per-foot GRFs, contact diagnostics, and replayable runs.
- Implementing quasi-random morphology and marker-layout sampling across male and female humanoid models to study tracking and contact robustness.
- Added reproducible CMake builds plus contract and integration tests for simulation and run-bundle invariants.